

aflevering_2

October 27, 2025

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[148]: import numpy as np
import matplotlib.pyplot as plt

plt.rcParams.update({
    "font.size": 14,          # default text size
    "axes.labelsize": 14,     # x/y labels
    "axes.labelweight": "normal",
    "xtick.labelsize": 10,     # x tick labels
    "ytick.labelsize": 10,     # y tick labels
    "legend.fontsize": 10,     # legend text
    "lines.linewidth": 1,      # thicker plot lines
    "grid.linewidth": 1.1
})

[65]: measurement_data = np.loadtxt("data.txt")

unique_concentration_values = np.unique(measurement_data[:,0])
data_groups = {val: measurement_data[measurement_data[:,0] == val] for val in unique_concentration_values}

[169]: D = 53 * 10**-3
markers = ['o', 's', '^', 'D', 'v', '*']

fig, ax1 = plt.subplots()

ax1.set_xlabel(r"$\Pi_2^{-1} = \mathrm{\frac{\rho}{\omega d^2}}$ [Re]")
ax1.set_ylabel(r"$\Pi_1 = \mathrm{\frac{P}{d^5 \omega^3}}$ [Re]")
ax1.set_xscale("log")

for i, val in enumerate(unique_concentration_values):
    data = data_groups[val]
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sucrose_concentration, rho, mu, rpm, torque = data[:,0], data[:,1], data[:,
↪ 2], data[:,3], data[:,4]

angular_frequency = rpm * 2 * np.pi / 60

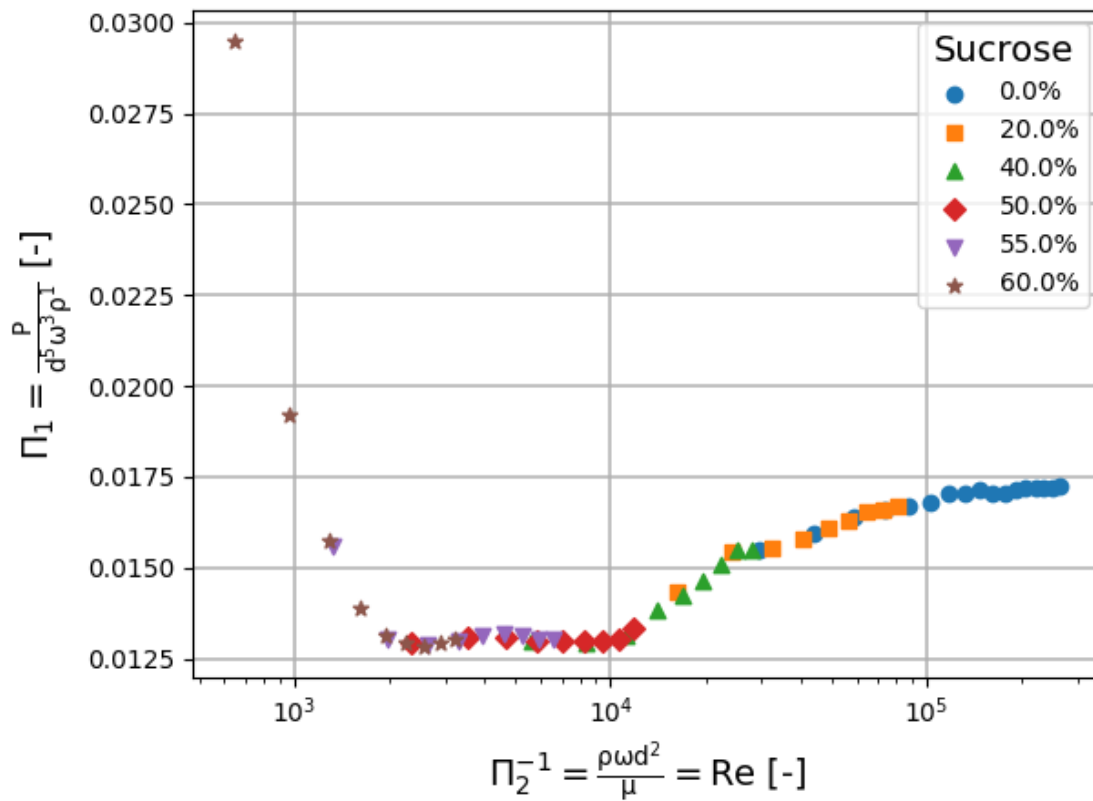
PI_1 = torque * angular_frequency / D**5 / angular_frequency**3 / rho

PI_2 = mu / D**2 / angular_frequency / rho

ax1.scatter(PI_2**-1, PI_1, marker=markers[i], label=f"{val}%", zorder=3)

ax1.grid()
ax1.legend(title="Sucrose")
plt.tight_layout()
plt.savefig("Pipilot.pdf", dpi=300)

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[168]: fig, ax1 = plt.subplots()

ax1.plot(unique_concentration_values, [data_groups[var][:, 2][0] for var in_
↪ unique_concentration_values], "bo-", label="mu")

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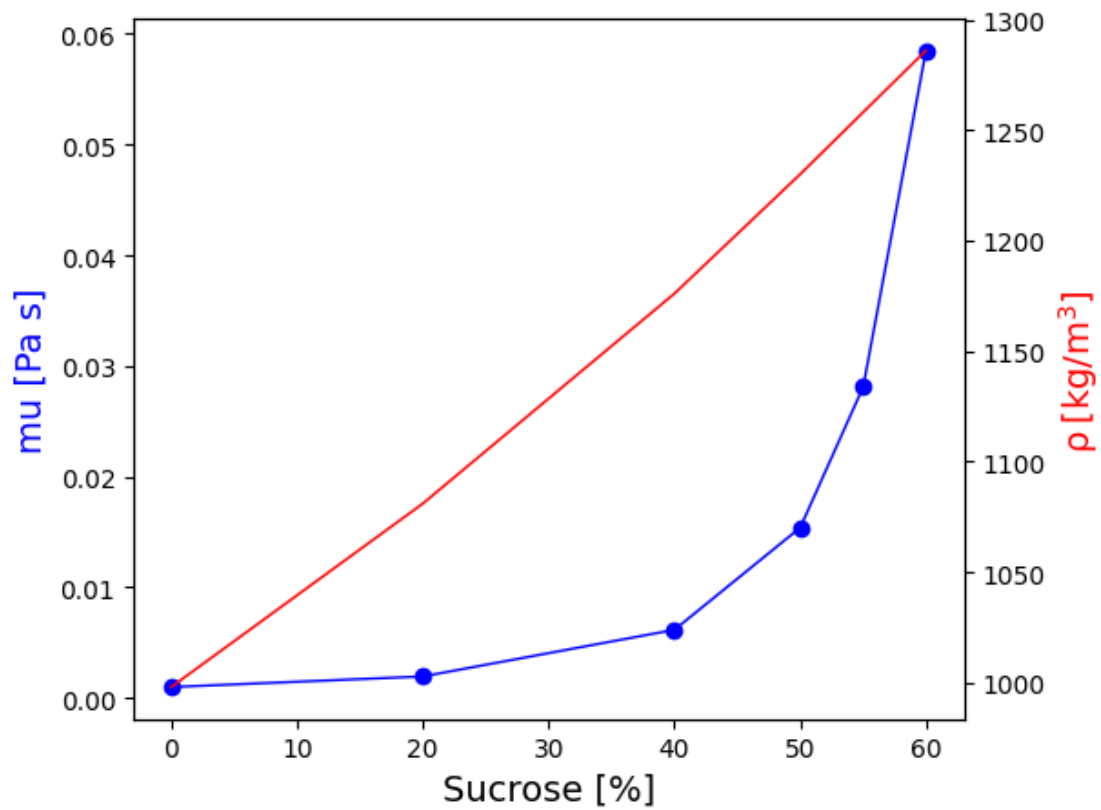
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ax1.set_xlabel("Sucrose [%]")
ax1.set_ylabel("mu [Pa s]", color="b")
ax2 = ax1.twinx()

ax2.plot(unique_concentration_values, [data_groups[var][:, 1][0] for var in unique_concentration_values], label="rho", color="r")
ax2.set_ylabel(r"$\mathrm{\rho} \text{ : [kg / m^3]}$", color="r")

plt.tight_layout()
plt.savefig("sucrose.pdf", dpi=300)

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[]: